FORM FOUR PRE- NATIONAL EXAMINATION
041 BASIC MATHEMATICS
MARKING SCHEME 2023

1. (a) Find the LCM 5 and 9

3	5	9
3	5	3
5	5	1
	1	1

.....01 mark

LCM= $3 \times 3 \times 5$.....01 mark

LCM $= 45$

Therefore the 45th person will be the first person to receive both passes.....01 mark

(b)

	Number	Type	Decimal form	Types of decimal
SN	$\frac{1}{5}$	Rational	0.2	Terminating
1	$\frac{5}{13}$	Rational	$0.\dot{3}\,84615\dot{}$	Recurring or Repeating
2	$\sqrt{9}$	Rational	3.0	Terminating
3	$\sqrt{13}$	Irrational	3.6056	Non recurring nor non terminating

.....@00 $\frac{1}{3}$ mark

2. (a) Let the number raised be y

$y^5 = 16\,807$ .....01 mark

$y = \sqrt[5]{16\,807}$.....01 mark

$$y = 7$$

Therefore the number is 7 .....01 mark

(b)$\log 25 + 2\log x = 2$

$$\log 25 + \log x^2 = 2$$
$$\log(25 \times x^2) = 2$$

$25x^2 = 10^2$.....01 mark

$$25x^2 = 100$$
$$\frac{25x^2}{25} = \frac{100}{25}$$

$x^2 = 4$.....01 mark

$$\sqrt{x^2} = \pm\sqrt{4}$$

$x = \pm 2$ .....01 mark

3. (a)Let the number of tourists who went to Manyara National Park be M

Let the number of tourists went Mikumi National Park be$K$

$$n(M) = 17, n(K) = 13, n(M \cap K)' = 8, n(\mu) = 29$$
$$n(M \cup K) = n(\mu) - n(M \cap K)'$$

$$= 29 - 8$$

$= 21$ ……..01 mark

Again $n(M \cup K) = n(M) + n(K) - n(M \cap K)'$

$$21 = 17 + 13 - n(M \cap K)'$$

$n(M \cap K)' = 30 - 21$……01 mark

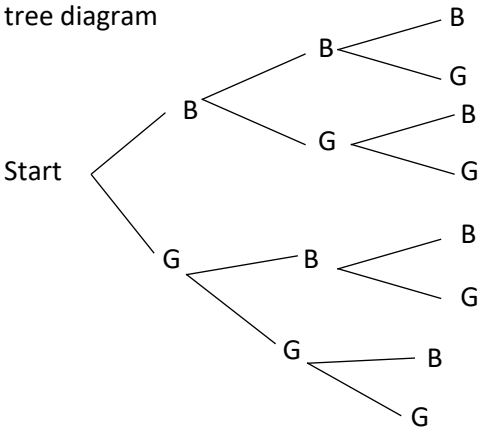
$$= 9$$

Therefore 9 tourists went to both National Parks……..01 mark

(b) Let B stands for boys children

Let G stands for girls children

By tree diagram



Sample space(S)= $\{BBB\ , BBG\ , BGB,\ BGG, GBG, GGB, GGG\ GBB\}$

$n(S) = 8$ ……..01 mark

i. Event$(E) = \{BBB, BGB, BGG, BBG\}$

$$n(E) = 4$$

$P(E) = \frac{n(E)}{n(S)} = \frac{4}{8} = \frac{1}{2} or\ 0.5$……01 mark

ii. Event$(E) = \{BBG, BGG\}$

$$n(E) = 2$$

$P(E) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4} = 0.25$ ……01 mark

4. (a)Express $6x - 4y - 3 = 0$ in form $y = mx + c$ or solve for y

$$4y = 6x - 3$$

$$\frac{4y}{4} = \frac{6x - 3}{4}$$

$$y = \frac{3x}{2} - \frac{3}{4}$$

$m_1 = \frac{3}{2}$……01 mark

Condition for parallel lines $m_1 = m_2 = \frac{3}{2}$

$\frac{3}{2} = \frac{8-5}{3-t}, \quad 6 = 9 - 3t$

$6 - 9 = -3t\ , \quad -3 = -3t,\ \frac{-3t}{-3} = \frac{-3}{-3}\ , t = 1$……01 mark

$$m = \frac{3}{2}, (1,5), (x,y)$$

$$\frac{3}{2} = \frac{y-5}{x-1}\ , 3x - 3 = 2y - 10$$

$$3x = 3 + 10 = 2y - 10 + 10$$

$$3x + 7 = 2y, \qquad \frac{3x}{2} + \frac{7}{2} = \frac{2y}{2}$$

$y = \frac{3x}{2} + \frac{7}{2}$……01 mark

(b) i. $2\underline{a} + \underline{b} - 3\underline{c} = 2\begin{pmatrix} 6 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 3\begin{pmatrix} 3 \\ -5 \end{pmatrix}$

$= \begin{pmatrix} 12 \\ -4 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -9 \\ 15 \end{pmatrix}$……01 mark

$= \begin{pmatrix} 12 + 2 - 9 \\ -4 + 1 + 15 \end{pmatrix}$

$= \begin{pmatrix} 5 \\ 12 \end{pmatrix}$……01 mark

Therefore $2\underline{a} + \underline{b} - 3\underline{c} = 5\underline{i} + 12\underline{j}$……01 mark

ii. $t\begin{pmatrix} 6 \\ -2 \end{pmatrix} + k\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$

$6t + 2k = 4$……equation 1

$-2t + k = -3$ equation 2 ……01 mark

Solve equation 1 and equation 2 by elimination method

$$6t + 2k = 4 \; is \; the \; same \; as \; 3t - k = 2$$

$$\begin{cases} 3t + k = 2 \\ -2t + k = -3 \end{cases}$$

$$- \; \Big| \begin{matrix} 3t + k = 2 \\ -2t + k = -3 \end{matrix}$$

$5t = 5, \frac{5t}{5} = \frac{5}{5}, t = 1$……01 mark

Substitute 1 for t in equation 2

$-2 \times 1 + k = -3, \; k = -3 + 2, \; k = -1$……01 mark

5. (a) Total parts $= 10 + 9 + 5 = 24 \; parts$

The sum of exterior angles of any polygon is 360°

1st exterior angle $= \frac{10}{24} \times 360^\circ = 150^\circ$……$00\frac{1}{2}$ mark

2nd exterior angle $= \frac{9}{24} \times 360^\circ = 135^\circ$……$00\frac{1}{2}$ mark

3rd exterior angle $= \frac{5}{24} \times 360^\circ = 75^\circ$……$00\frac{1}{2}$ mark

Interior angle + exterior angle $= 180^\circ$

1st interior angle$= 180^\circ -$exterior angle$180^\circ - 150^\circ = 30^\circ$……$00\frac{1}{2}$mark

2nd interior angle$= 180^\circ -$exterior angle $180^\circ - 135^\circ = 45^\circ$……$00\frac{1}{2}$ mark

3rd interior angle$= 180^\circ -$exterior angle $180^\circ - 75^\circ = 105^\circ$

Therefore the interior angles are $30^\circ, 45^\circ \; and, 105^\circ$……$00\frac{1}{2}$ mark

(b) $m \angle BEA \cong m \angle AEB$ (Common angle (A))……01 mark

$\overline{BE} \cong \overline{AE}$ (Given side (S)) ……$00\frac{1}{2}$mark

$m \angle DAE \cong \; m \angle CBE$ (Given angle (A))…… $00\frac{1}{2}$ mark

Therefore $\Delta \, BCE \cong \Delta ADE$ (By ASA postulate)……01 mark

6. (a) i. $y \propto \sqrt[3]{x_1}$ ……01 mark

$$y = k\sqrt[3]{x_1}, x_1 = 27, y_1 = 9, x_2 = 8, required \; y_2$$

$$k = \frac{y}{\sqrt[3]{x_1}}$$

$$= \frac{9}{\sqrt[3]{27}} = \frac{9}{3}$$

$k = 3$……01 mark

$$3 = \frac{y}{\sqrt[3]{8}}, = 2 \times 3$$

$6 = y_2$……01 mark

(b) Let m , h , d represents the number of men , hours, and days respectively

$m_1 = 40$, $h_1 = 8$, $d_1 = 16, m_2 = 16, h_2 = 10$, $required\ d_2$

$$m_1 \propto \frac{1}{h_1 d_1}, \quad m_1 = \frac{k}{h_1 d_1}$$

$k = m_1 h_1 d_1$ ……01 mark

$k = 40 \times 10 \times 16$ ……01 mark

$$k = m_2 \times h_2 \times d_2$$

$$40 \times 8 \times 16 = 16 \times 10 \times d_2$$

$$d_2 = \frac{40 \times 8 \times 16}{16 \times 10}$$

$= 32$ ……01 mark

7. (a) Let the buying price be y

Percentage profit= $\left(\frac{selling\ price - buying\ price}{buying\ price}\right) 100\%$……$00\frac{1}{2}$ mark

$$15\% = \left(\frac{34\ 5000 - y}{y}\right) 100\%$$

$$\frac{15\%}{100\%} = \frac{34\ 500 - y}{y} \times \frac{100\%}{100\%}$$

$$0.15 = \frac{34\ \ 500 - y}{y}$$

$$0.15y = 34\ 500 - y$$

$$0.15y + y = 34\ 500 - y + y$$

$1.15y = 34\ 500$ ……$00\frac{1}{2}$ mark

$$\frac{1.15y}{1.15} = \frac{34\ 500}{1.15}$$

$y = 30\ 000$……01 mark

For five text books $= 30\ 000 \times 5 = 150\ 000$

Therefore the buying price of 5 text books is 150 000 shillings ……01 mark

Alternatively

$115\% = 34\ 5000$

$100\% = y$ ……$00\frac{1}{2}$ mark

$y = \frac{100\% \times 34\,500}{1155}$ ……$00\frac{1}{2}$mark

$y = 30\,000$ ……01 mark

For five text books $= 30\,000 \times 5 = 150\,000$

Therefore the buying price of 5 text books is 150 000 shillings ……01 mark

(b) PROFIT AND LOSS ACCOUNT FOR THE YEAR ENDED 31st DECEMBER 2019

Dr Cr

Particulars	Folio	Amount	Particulars	Folio	Amount
Wage and salaries		100 000	Gross profit	b/d	900 000
Rent		50 000	Discount received		80 000
Insurance		120 000	Commission received		400 000
Discount allowed		300 000			
Transport		130 000			
Office expenses		200 000			
Bad debts		20 000			
Advertising		70 000			
Sundry expenses		150 000			
Net profit		240 000			
		1 380 000			**1 380 000**

8. (a) $A_1 = 73\,, A_2 = 78, A_n = 588$

$$d = A_2 - A_1$$
$$= 78 - 75$$

$= 5$ ……01 mark

$$A_n = A_1 + (n-1)d$$
$$588 = 73 + (n-1)5$$
$$588 = 73 + 5n - 5$$
$$588 = 68 + 5n$$
$$588 - 68 = 68 - 68 + 5n$$

$520 = 5n$ ……$00\frac{1}{2}$ mark

$$\frac{520}{5} = \frac{5n}{5}$$

$n = 104$ ……$00\frac{1}{2}$mark

$$S_n = \frac{n}{2}(A_1 + A_n)$$
$$= \frac{104}{2}(73 + 588)$$
$$= 52(661)$$

$= 34\,372$ ……01 mark

(b)$G_1 = x - 1\,, G_2 = x,\ G_3 = x = 2$

$$\frac{G_2}{G_1} = \frac{G_3}{G_2} = r$$

$\frac{x}{x-1} = \frac{x+2}{x}$……$00\frac{1}{2}$ mark

$$x^2 = x^2 + 2x - x - 2$$
$$0 = x - 2$$

$x = 2$……01 mark

$G_1 = x - 1 = 2 - 1 = 1, \quad G_2 = x = 2\,, \quad G_3 = x + 2 = 2 + 2 = 4$

The sequence is 1, 2, 4,

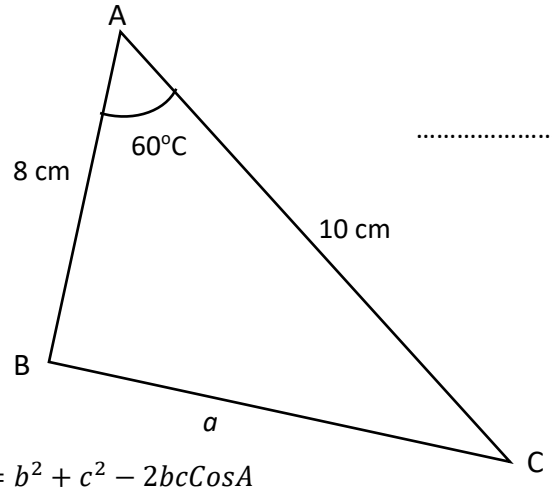
$$G_{15} = G_1 r^{n-1}$$

$= 1 \times 2^{15-1}$……$00\frac{1}{2}$ mark.

$$= 2^{14}$$

$= 16\,384$……01 mark

9. (a) i. According to the given data Side– Side –Side (SAS) is used hence'a' is calculated by using cosine rule. ……01 mark



……$00\frac{1}{2}$ mark.

$a^2 = b^2 + c^2 - 2bcCosA$

$$a^2 = b^2 + c^2 - 2 \times 8 \times 10Cos60^{\circ}$$
$$= 64 + 100 - 160 \times 0.5$$
$$= 164 - 80$$
$$= 84$$
$$a^2 = 84\,, \sqrt{a^2} = \pm\sqrt{84}$$
$$a = \pm 9.2\ cm$$

$a = 9.2\ cn\,, a \geq 0$ (2 decimal places)……01 mark

(b) $c^2 = a^2 + b^2$

$a^2 = 25^2 - 20^2 = 625 - 400 = 225$ ……01 mark

$a^2 = 225$ ……01 mark

$$\sqrt{a^2} = \pm\sqrt{225}$$
$$a = \pm 15, a = 15, a \geq 0$$

Therefore the other side is 15 cm……01 mark

10. (a) $10\left(\frac{m}{2} - \frac{n}{5} = 1\right)$ is the same as $5m - 2n = 10$ ……equation1

$3n - m = 24$ ……equation 2

Solve equation 1 and 2 by elimination method

$\begin{cases} 5m - 2n = 10 \\ -m + 3n = 24 \end{cases}$……01 mark

$$\left|\begin{matrix} 5m - 2n = 10 \\ -m \times 5 + 5 \times 3n = 5 \times 24 \end{matrix}\right.$$

$$-\left|\begin{matrix} 5m - 2n = 10 \\ -5m + 15n = 120 \end{matrix}\right.$$

$$13n = 130$$

$\frac{13n}{13} = \frac{130}{13}, n = 10$ ……01 mark

Substitute 10 for n in equation 2

$3 \times 10 - m = 24\ m = 30 - 24 = 6$ ……01 mark

$\frac{x+1+x}{x(x+1)} = \frac{9}{20}$

$\frac{2x+1}{x(x+1)} = \frac{9}{20}$

$$20(2x+1) = 9x(x+1)$$
$$40x + 20 = 9x^2 + 9x$$
$$9x^2 + 9x - 40x - 20 = 0$$
$$9x^2 - 31x - 20 = 0$$
$$\frac{9x^2}{9} - \frac{31x}{9} - \frac{20}{9} = \frac{0}{9}$$
$$x^2 - \frac{31}{9}x - \frac{20}{9} + \frac{20}{9} = \frac{20}{9}$$

$x^2 - \frac{31}{9}x = \frac{20}{9}$ ……01 mark

$$x^2 - \frac{31}{18} + \frac{961}{324} = \frac{20}{9} + \frac{961}{324}$$
$$\left(x - \frac{31}{18}\right)\left(x - \frac{31}{18}\right) = \frac{720 + 961}{324}$$

$\left(x - \frac{31}{18}\right)^2 = \frac{1681}{324}$……01 mark

$$\sqrt{\left(x - \frac{31}{18}\right)^2} = \pm\sqrt{\frac{1683}{324}}$$
$$x - \frac{31}{18} = \pm\frac{41}{18}$$
$$x - \frac{31}{18} + \frac{31}{18} = \frac{31}{18} \pm \frac{41}{18}$$
$$x = \frac{31+41}{18}\ or\ x = \frac{31-41}{18}\quad x = \frac{72}{18}\ or\ \frac{-10}{18}$$

$x = 4\ or\ x = \frac{-5}{9}$ ……01 mark

11. (a) Cumulative frequency table

Upper class boundary	Cumulative frequency	Class mark	f	Class interval
10	2	5	2	0–10
20	6	15	4	10–20
30	8	25	2	$20-30$
40	16	35	8	$30-40$
50	22	45	6	$40-50$
60	24	55	2	$50-60$
Total			24	

Hint: upper class limit are exclusive

From the Histogram

Mode score is about 37.5

Median class is $30-40$

Median score$= l + \left[\frac{\frac{N}{2} - fb}{fw}\right] i$

$$= 30 + \left[\frac{\frac{24}{2} - 8}{8}\right] 10$$

$$= 30 + 5$$

Median score $= 35$

(b) $m\angle A = 90^{\circ}$(Angle in a triangle at the circumference subtended by a semi- circle equals right angle)……01 mark

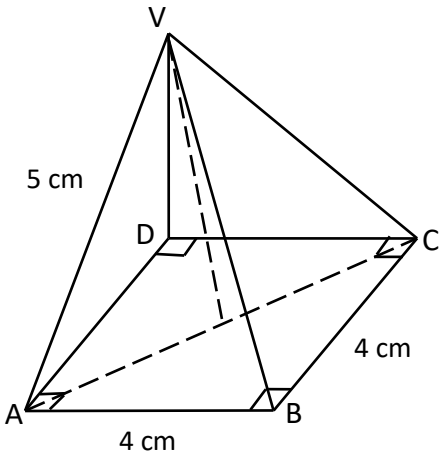
$32^{\circ} + 90^{\circ} + t = 90^{\circ}$ (Sum of the interior angle in a triangle)……01 mark

$122^{\circ} + t = 180^{\circ}$……01 mark

$$t = 180^{\circ} - 122^{\circ}$$

$t = 58^{\circ}$……01 mark

12. (a)



Applying Pythagoras theorem in ΔABC

$$\left(\overline{AC}\right)^2 = \left(\overline{BC}\right)^2 + \left(\overline{AB}\right)^2$$

$$\left(\overline{AC}\right)^2 = 4^2 + 4^2$$

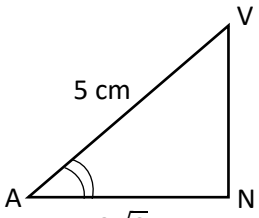
$\left(\overline{AC}\right)^2 = 32$ ……01 mark

$$\sqrt{\left(\overline{AC}\right)^2} = \sqrt{32}$$

$$= 4\sqrt{2}$$

$\overline{AC} = \left(4\sqrt{2}\right) cm$, $\overline{AC} \geq 0$ ……01 mark

i. From ΔANV



……$00\frac{1}{2}$ mark.

$\overline{AN} = \frac{1}{2} \times \overline{AC} = \frac{1}{2} \times 4\sqrt{2} = 2\sqrt{2}$……01 mark

Cos A $= \frac{2\sqrt{2}}{5}$ 0.5657 $A = cos^{-1}(0.5657) = 55^{\circ}33'$

Therefore the angle between VA and the base ABCD is $55^{\circ}33'$ .....01 mark

(b) Angular distance or angle subtended between points R and Q

$$= 28^{\circ} + 72^{\circ}$$

$= 100^{\circ}$.....01 mark

Distance $\overline{PQ}$ is on latitude or small circle$30^{\circ}N$

$$l = \frac{\pi r\theta}{180^{\circ}} \; but \; r = Rcos\alpha$$

$$l = \frac{\pi R\theta Cos\alpha}{180^{\circ}}$$

$= \frac{3.142\times6370\ km\times100^{\circ}\times cos30^{\circ}}{180^{\circ}}$.....01 mark

$$= \frac{1733310.009}{180}$$

$= 9629.50\ km$.....01 mark

$$t = \frac{D}{v}$$

$= \frac{9629.50\ km\ h}{650\ km}$.....01 mark

$= 15\ hours$.....01 mark

13. (a) $y = x, m = 1\ tan\theta = 1\ tan^{-1}(1) = 45^{\circ}$.....$00\frac{1}{2}$ mark.

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = M_x = y\begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} cos2\theta & sin2\theta \\ sin2\theta & -cos2\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

$$= \begin{bmatrix} cos90^{\circ} & sin90^{\circ} \\ sin90^{\circ} & -cos90^{\circ} \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 4 \\ 2 \end{bmatrix}$.....$00\frac{1}{2}$ mark.

$= \begin{bmatrix} 2 \\ 4 \end{bmatrix}$.....$00\frac{1}{2}$ mark.

Matrix of rotation for angle θ is

$R_{\theta} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$ Since rotation is 90° clockwise $\theta = -90^{\circ}$

$$R = \begin{bmatrix} \cos(-90^{\circ} & -\sin(90^{\circ} \\ \sin(-90^{\circ} & \cos(-90^{\circ} \end{bmatrix}$$

$R = \begin{bmatrix} cos90^{\circ} & sin90^{\circ} \\ -sin90^{\circ} & cos90^{\circ} \end{bmatrix}$, but $sin(-90^{\circ}) = -sin90^{\circ}$ and $\cos(-90^{\circ}) = \cos(90^{\circ})$

$= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.....01 mark

Thus the image of point (2,4) under the rotation -90° is

$$\begin{bmatrix} x'' \\ y'' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$= \begin{bmatrix} 4 \\ -2 \end{bmatrix}$.....01 mark

(b) (i) $\begin{cases} 2x+3y=-5 \\ 5x-y=13 \end{cases}$ by equating corresponding elements

$\begin{cases} 2x+3y=-5 \\ 5x-y=13 \end{cases}$ ……Solve equation 1 and 2Type equation here.

$$\begin{vmatrix} 2x+3y=-5 \\ 3\times 5x-3\times y=3\times 13 \end{vmatrix} = \begin{cases} 15x-3y=39 \\ 2x+3y=-5 \end{cases}$$

$$+\begin{vmatrix} 15x-3y=39 \\ 2x+3y=-5 \end{vmatrix}$$

$$17x=34$$

$$\frac{17x}{17}=\frac{34}{17}$$

$x=2$……01 mark

Substitute 2 for$x$ in equation 2

$$5\times 2-y=13$$
$$y=13-10$$

y = 3……01 mark

(ii) $\begin{cases} 5x-y=12 \\ 7x+y=12 \end{cases}$

Express the equation in matrix form

$$\begin{bmatrix} 5 & -1 \\ 7 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 12 \\ 12 \end{bmatrix}$$

Let $\begin{bmatrix} 5 & -1 \\ 7 & 1 \end{bmatrix}$ be A

$|A|=(5+7)=12$ ……..01 mark

$A^{-1}=\frac{1}{12}\begin{bmatrix} 1 & 1 \\ -7 & 5 \end{bmatrix}=\begin{bmatrix} \frac{1}{12} & \frac{1}{12} \\ \frac{-7}{12} & \frac{5}{12} \end{bmatrix}$……01 mark

$$AA^{-1}=I$$

$$\begin{bmatrix} 5 & -1 \\ 7 & 1 \end{bmatrix}\begin{bmatrix} \frac{1}{12} & \frac{1}{12} \\ \frac{-7}{12} & \frac{5}{12} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} \frac{1}{12} & \frac{1}{12} \\ \frac{-7}{12} & \frac{5}{12} \end{bmatrix}\begin{bmatrix} 12 \\ 12 \end{bmatrix}$$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} \frac{1}{12}\times 12+\frac{1}{12}\times 12 \\ \frac{-7}{12}\times 12+\frac{5}{12}\times 12 \end{bmatrix}$……01 mark

$$\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 1+1 \\ -7+5 \end{bmatrix}$$

$\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} 2 \\ -2 \end{bmatrix}$……01 mark

14. (a)Summarize the information in the table as follows:

	Tables	Chairs	Cost
Plant A	20	60	\$1 000
Plant B	25	50	\$ 9000
Requirements	200	500	

Let the number of days to operate plant A be x

Let the number of days to operate plant B be$y$

$$20x + 25y \geq 200 \; is \; the \; as \; \frac{20}{5}x + \frac{25}{5}y \geq \frac{200}{5} = 4x + 5y \geq 40$$

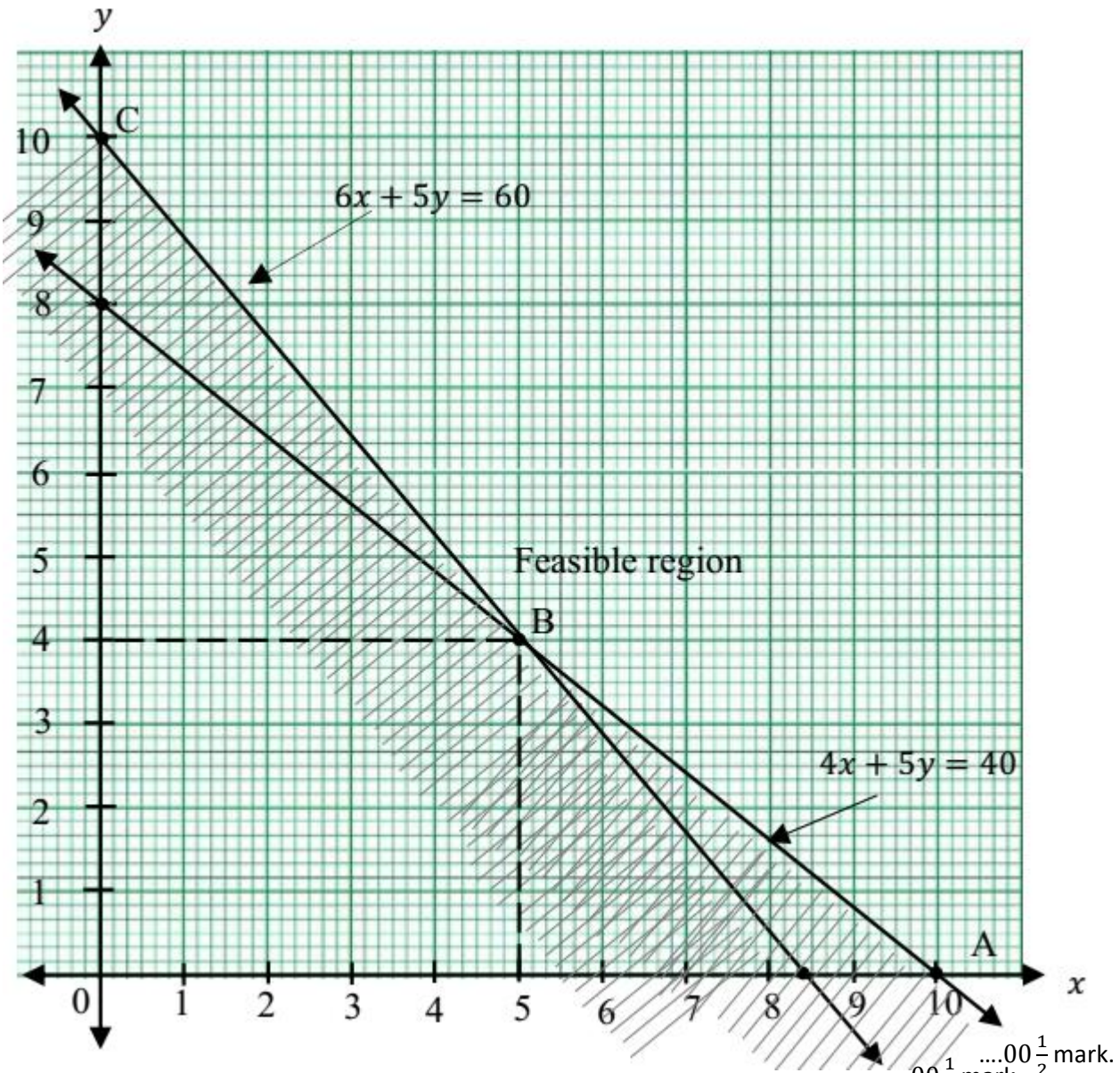
Minimize C $= \$1\,000\text{x} + \$900\text{y}$……01 mark

Subject to $\begin{cases} 4x + 5y \geq 40 \\ 6x + 5y \geq 50 \\ x \geq 0\,, y \geq 0 \end{cases}$……01 mark

By $x - ntercep \; and \; y - intercept$

$4x + 5y \geq 40$ i.e(10, 0),(0 , 8)

$6x + 5y \geq 50$ i.e $(\frac{25}{3},\ 0)$,(0, 10)



….$00\frac{1}{2}$ mark.

……$00\frac{1}{2}$ mark.

Points	$f(x,y) = \$1\,000x + \$900y$	Value
A (10 , 0)	$10 \times \$1\,000 + 0 \times \$\,900$	$10 000
B (5,4)	$5 \times \$\,1\,000 + 4 \times \$\,900$	$\$\,8\,600$
C(0, 10)	$0 \times \$\,1\,000 + 10 \times \$900$	$\$\,9\,000$

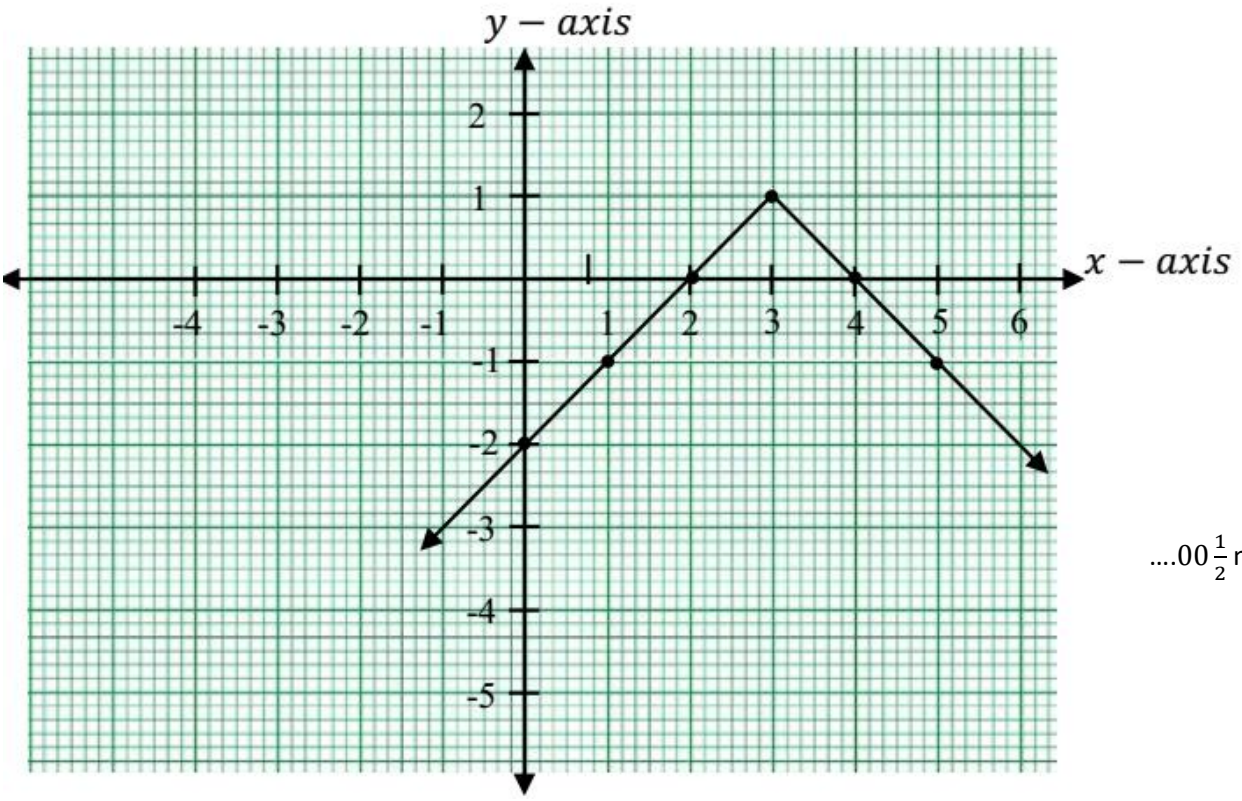
……$01\frac{1}{2}$ marks

i. Plant A should operate 5 days and 4 days to operate plant B

ii. Minimum cost is $\$8\,600$……01 mark

(b) Table of values $f(x) = y = -|x-3| + 1$

	-2	-1	0	1	2	3	4	5	6
y	-4	-3	-2	-1	0	1	0	-1	-2



….$00\frac{1}{2}$ mark.

Domain $= \{x: x\varepsilon\mathbb{R}\}$ ……$00\frac{1}{2}$mark

Range $= \{y: y \leq 1\}$……$\mathbf{00\frac{1}{2}}$ mark